

Bhanuprakash Bhat

Cloud Data Engineer | AWS Certified AI Practitioner | Microsoft Certified Power BI Data Analyst | Microsoft Certified Azure Data Fundamentals

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PROFILE

3x Cloud-Certified Data Engineer (AWS AI Practitioner, MS Power BI Data Analyst, Azure Data Fundamentals) with hands-on experience designing, deploying, and maintaining production-grade data pipelines and ML systems across GCP, Azure, and AWS. Proven ability to architect end-to-end solutions—from serverless cloud pipelines and ETL workflows to autonomous AI agents—delivering measurable business impact including 100% automation of manual triage processes and projected \$50K+ annual savings in consulting costs. Currently completing a Master's in Data Science & AI at the University of Sydney.

TECHNICAL SKILLS SUMMARY

- Languages & Databases::** Python, SQL, R, Java, PostgreSQL, MySQL, DuckDB
- Data Engineering::** Apache Spark, Databricks, dbt Core, ETL/ELT Pipelines, Data Modelling, Supabase, ChromaDB, Vector Databases
- Cloud Platforms::** GCP (BigQuery, Cloud Run, Terraform), Azure (Data Factory, Synapse Analytics, Databricks), AWS
- AI / ML Frameworks::** Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, LangChain, LangGraph, Hugging Face, OpenCV
- Data Analysis & BI::** Power BI (DAX, Power Query), Excel (Pivot Tables, Power Pivot), Seaborn, Matplotlib, Pandas, NumPy
- DevOps & Tools::** Docker, Git, GitHub, FastAPI, RESTful APIs, CI/CD, Streamlit, Jupyter, VS Code

PROFESSIONAL EXPERIENCE

- Cloud Data Engineer** 01/2026 | Sydney, Australia
Demografy 🔗
- Designed and implemented scalable data pipelines integrating Google Places API with authentication, smart caching, and cost monitoring, reducing API overhead and optimizing cloud resource usage.
 - Built cloud-native database architecture on Supabase and PostgreSQL, designing schemas for raw JSON ingestion and suburb-level real estate metrics with indexing strategies for high-performance geospatial lookups.
 - Developed modular dbt Core transformation models (staging, intermediate, and mart layers) to convert raw API data into analytics-ready datasets, enabling real-time metric computation and BI reporting.
 - Deployed and maintained cloud-hosted data applications using Streamlit, implementing CI/CD-ready workflows and backend integration with sub-15-second processing time.
- Assessment and AI Transition Assistant** 05/2025 | Sydney, Australia
The University of Sydney
- Supported enterprise-wide transition to AI-enabled assessment systems, collaborating with Educational Designers and Unit Coordinators across curriculum platforms and policy-aligned workflows.
 - Validated, standardised, and governed assessment metadata within the Sydney Curriculum Management System, ensuring data integrity, schema consistency, and policy compliance.
 - Conducted assessment data quality audits, identifying structural inconsistencies, rule violations, and dependency issues across hundreds of unit outlines.
 - Contributed to responsible AI adoption and change management, supporting evaluation of emerging EdTech platforms and workflow automation capabilities.
- Data Engineer** 10/2024 – 03/2025 | Sydney, Australia
Clinic to Cloud 🔗
- Migrated patient records from on-premise clinical systems to Azure by designing end-to-end data pipelines using Azure Data Factory (ADF), landing raw CSV exports in Blob Storage and structuring a medallion architecture (bronze/silver/gold) in ADLS Gen2, reducing manual data handling by 80%

- Built and optimised ETL workflows in Azure Synapse Analytics using dedicated SQL pools and PySpark-based Spark pools for data cleansing, deduplication, and transformation of patient demographics, visit histories, and clinical interaction data, improving data accuracy.
- Consolidated transformed datasets into a Microsoft Fabric Lakehouse using notebooks and Dataflows Gen2, creating a unified analytics layer that replaced fragmented spreadsheet-based reporting and enabled seamless cross-functional data access for clinical and operational teams.
- Developed interactive Power BI dashboards connected to the Fabric Lakehouse for clinicians and operational staff, enabling real-time trend analysis of patient visits, longitudinal health indicators, and clinic performance metrics

Moog Inc01/2024 – 05/2024 | Bengaluru, India

Data Analyst Intern

- Designed and deployed dynamic Power BI dashboards to visualize critical sensor data (pressure, temperature, and density), enabling engineers and managers to track real-time trends and optimize system performance.
- Processed and cleaned large datasets using Pandas and NumPy, ensuring high data integrity before feeding into visualization pipelines and statistical analysis.
- Conducted exploratory data analysis (EDA) using Matplotlib and Seaborn to uncover hidden insights in system behavior, supporting early detection of anomalies in aerospace components.

PROJECTS

Serverless Autonomous AI Agent for Operational Triage | GCP, Docker, Cloud Run, OpenAI, Slack

- Architected a serverless pipeline on Google Cloud Run using containerised Python and Docker, eliminating idle compute costs through scheduled execution.
- Integrated OpenAI GPT-4o for semantic analysis on real-time email streams (IMAP), classifying urgency and routing actionable alerts to Slack via Webhooks with near-instant latency.
- Automated 100% of manual email triage, projecting ~\$10,600 in reclaimed annual productivity per employee.

Autonomous RAG Compliance Agent | LangChain, LangGraph, ChromaDB, Hybrid Search

- Engineered an autonomous agent to audit R&D Tax, EMDG, and ESIC claims, reducing preliminary legal assessment costs by ~95% (projected savings of \$50,000+ annually).
- Built a custom LangGraph workflow using Hybrid Search (Vector + Keyword) and ChromaDB, anchoring every decision in official legislation and eliminating hallucinations on critical financial caps.
- Developed a specialised Grader node to validate retrieval relevance, flagging 100% of tested Excluded Activities that typically trigger high-penalty regulatory audits.

Electricity Sector Data Integration & Spatial Analysis | DuckDB, Python, Geospatial

- Built an end-to-end data engineering pipeline to integrate national electricity, emissions, and ABS regional datasets.
- Performed data cleaning, augmentation, and geocoding to validate facility coordinates and handle missing or malformed geographic data.
- Designed and implemented a spatial database schema in DuckDB with normalized staging tables and a denormalized analytical mart supporting geometry types.

EDUCATION

Master of Technology in Data Science and Artificial Intelligence07/2024 – 06/2026 | Sydney, Australia

The University of Sydney

Bachelor of Technology in Aerospace Engineering08/2020 – 06/2024 | Bengaluru, India

M S Ramaiah University of Applied Sciences

PUBLICATIONS

Three-Dimensional Computational Study over a Hyper-Ballistic missile body at Hypersonic Mach numbers14/07/2023

ISSW-34, Daegu, South Korea